

# Max Rakitin

## Curriculum vitae

### About

Name: Max Rakitin (a.k.a. Maksim S. Rakitin)

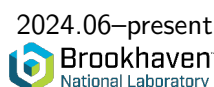
Summary: I am a group leader of the Data Acquisition and Detectors group of the Data Science and Systems Integration Division of NSLS-II, BNL.

News: "Computer, Is My Experiment Finished?" (September 16, 2022)  
<https://www.bnl.gov/newsroom/news.php?a=220832>

"Seeing the Forest Through the Trees: Brookhaven Lab Scientists Develop New Computational Approach to Reduce Noise in X-ray Data." (April 18, 2022)  
<https://www.bnl.gov/newsroom/news.php?a=219533>

Links: [BNL](#) • [SBU](#) • [SUSU](#)  
[@mrakitin](#) • [@mrakitin](#) • [Google Scholar](#) • [ResearchGate](#)  
[ORCID](#): 0000-0003-3685-852X

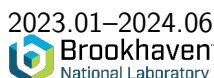
### Experience



2024.06–present

**Computational Scientist (RS-5), Group Leader**, *Data Science and Systems Integration (DSSI) Division, Data Acquisition and Detectors Group, NSLS-II, Brookhaven National Laboratory, Upton, NY* (<https://www.bnl.gov>, [DSSI](#)).

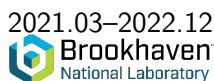
**Supervisor:** Dr. Stuart Wilkins



2023.01–2024.06

**Computational Scientist (RS-5), Deputy Group Leader**, *Data Science and Systems Integration (DSSI) Program, Data Acquisition and Detectors Group, NSLS-II, Brookhaven National Laboratory, Upton, NY* (<https://www.bnl.gov>, [DSSI](#)).

**Supervisor:** Dr. Thomas Caswell



2021.03–2022.12

**Associate Computational Scientist (RS-4), Supervisor**, *Data Science and Systems Integration (DSSI) Program, NSLS-II, Brookhaven National Laboratory, Upton, NY* (<https://www.bnl.gov>, [DSSI](#)).

**Supervisor:** Dr. Stuart Campbell



2019.10–2021.03

**Associate Computational Scientist**, *Data Acquisition, Management and Analysis (DAMA) Group, NSLS-II, Brookhaven National Laboratory, Upton, NY* (<https://www.bnl.gov>).

**Supervisor:** Dr. Stuart Campbell

2017.11–2019.09



**Assistant Computational Scientist, Data Acquisition, Management and Analysis (DAMA) Group, NSLS-II, Brookhaven National Laboratory, Upton, NY**  
(<https://www.bnl.gov>).

**Supervisor:** Dr. Stuart Campbell

#### Software Projects:

- **Bluesky** — a library for experiment control and collection of scientific data and metadata, [docs](#), [GitHub](#).
- **Ophyd** — a device abstraction library, [docs](#), [GitHub](#).
- **Databroker** — a simple, user-friendly interface for retrieving stored data and metadata from multiple sources, [docs](#), [GitHub](#).
- **Sirepo-Bluesky** — an interface library between the **Bluesky** data acquisition framework and the **Sirepo** browser-based interface framework to beamline and accelerator simulation codes, [docs](#), [GitHub](#).
- **Blop** — a Bayesian optimization library for autonomous experiment control, [docs](#), [GitHub](#).

#### Funded Projects (PI roles):

- **LDRD Type B “Simulation-aided Instrument Optimization Using Artificial Intelligence and Machine Learning Methods”** (LDRD #22-031, FY2022–FY2024, Oct. 2021–Sep. 2023) — PI: M. Rakitin, co-PIs: Y. Du, M. Fedurin. BNL Laboratory-Directed Research and Development project, <https://www.osti.gov/biblio/1972474>.
- **DOE SBIR “X-ray Beamline Control with an Online Model for Automated Tuning and Reconfiguration”** (Award No. DE-SC0020593, 2020–2023) — PI at BNL (subcontract). Lead organization: RadiaSoft, LLC (PI: B. Nash).
- **DOE SBIR “An Integrated Platform for High Performance Data Management and Analysis at X-ray Light Sources”** (Award No. DE-SC0021553, 2021–2024) — PI at BNL (subcontract). Lead organization: RadiaSoft, LLC (PI: N. Cook).

#### Responsibilities:

- Scientific software development & maintenance.
- Packaging and maintenance of conda packages (conda-forge, and formerly lightsource2-tag and nsls2forge conda channels).
- Deployment of the software with Ansible.
- NSLS-II beamlines and users support.
- Active collaboration with peers at other US DOE National Labs’ User Facilities (SLAC — LCLS & SSRL, LBNL — ALS, ANL — APS, ORNL — SNS) and international facilities (DLS, PSI, BESSY-II, CLS, DESY).
- Teaching people to use our software stack.
- Mentor to student interns (summer 2016: 1 intern, summer 2018: 1 intern, summer 2019: 1 intern, spring and summer 2020: 2 interns, summer 2021: 2 interns, summer 2023: 1 intern, summer 2024: co-mentor to 1 intern).

2015.12–2017.10



**Research Associate (Postdoc), NSLS-II, Brookhaven National Laboratory, Upton, NY**  
(<https://www.bnl.gov>).

**Supervisor:** Dr. Oleg Tchoubar (Chubar)

**Software Projects:**

- **Synchrotron Radiation Workshop (SRW)** — computer code for X-ray source and optics simulations, <https://github.com/ochubar/SRW>.
- **Sirepo** — a cloud-based framework for SRW, <https://github.com/radiasoft/sirepo>.
- **Image processing and data visualization**, <https://github.com/mrakitin/plotting> and <https://github.com/mrakitin/experiments>.
- **CRL simulator** — a code for simulation of a translocator (compound refractive lenses (CRL) for X-ray focusing), <https://github.com/mrakitin/bnlcrl>.

**Responsibilities:**

- Adding new functionality to Sirepo (e.g., new optical elements, new reports, [implementation](#) of dynamical access of crystal data and optical constants from external servers).
- Adding new functionality to SRW (e.g., new optical elements).
- Creation of new SRW/Sirepo “virtual beamline” scripts/examples.
- Creation and update of [Sirepo](#) & [SRW](#) wiki documentation.
- SRW & Sirepo users community support via GitHub issues, email communication, *etc.*
- Deployment of SRW & Sirepo to NSLS-II servers for usage by beamline scientists.
- Computational support for NSLS-II beamlines — carrying out required SRW & Sirepo simulations.
- Assistance with scan plans preparation using **Bluesky** data collection framework and **ophyd** package for controlling motors, detectors, *etc.*
- Participation in X-ray experiments (SRX, ESM, SMI, CHX beamlines of NSLS-II) — carrying out measurements and bulk data analysis, processing & visualization using Python (NumPy, SciPy, Matplotlib, PIL, *etc.*; raw data and images, NumPy arrays, HDF5 datasets).
- Implementation and integration of samples simulation code for coherent scattering experiments in SRW and Sirepo (collaborative project with CFN).
- Assisting with development of the Hartmann mask optical element in SRW and its implementation in Sirepo (collaborative project with the Metrology group of NSLS-II).
- Tutorship of summer students.

2013.10–2015.12



**Postdoctoral Associate (Postdoc)**, *Prof. Oganov's lab, Department of Geosciences, Stony Brook University, Stony Brook, NY* (<https://stonybrook.edu>, <https://uspex-team.org/en/uspex>).

**Supervisor:** Prof. Artem R. Oganov

**Software Projects:**

- Refactoring of [USPEX](#) code and porting it from Matlab to Python using modern programming techniques.
- *Ab initio* investigation of BeF<sub>2</sub> and SiO<sub>2</sub> systems in a wide pressures range ([paper](#)) using [USPEX](#), [VASP](#), [Quantum Espresso](#), [Phonopy](#).
- Prediction of secondary structures of proteins from knowledge of sequences of amino acids. Development of USPEX interface for [Tinker](#).

**Responsibilities:**

- Main developer of the project on rewriting **USPEX** in Python with use of NumPy, SciPy, TkInter, etc. Web-based API creation.
- New **releases** of Matlab/Octave version (USPEX 9.4.1, 9.4.2, 9.4.3, 9.4.4), new features, bug fixes, maintenance.
- Development and maintenance of USPEX continuous integration system (automatic nightly builds + testing).
- Development of tests for USPEX interface with VASP, GULP, LAMMPS, CASTEP, Quantum Espresso, SIESTA, CP2K, QuantumWise ATK, DMACRYS, Tinker, FHI-aims.
- Development and maintenance of online utilities <https://uspex-team.org> using JavaScript, jQuery, PHP, HTML, JSON, REST API, WebGL, XML, SVN, interfaces to Python, Fortran, Matlab programs, etc.
- Installation, configuration, and maintenance of an instance of a Trac system and SVN.
- Creation and maintenance of **USPEX manual** in  $\text{\LaTeX}$ , PDF, HTML formats.
- **USPEX community** support.

2008.10–2013.09 **Researcher**, *Department of General and Theoretical Physics, South Ural State University, Chelyabinsk, Russia* (<https://www.susu.ru/en>).



**Supervisor:** Prof. Alexander A. Mirzoev

**Software Projects:**

- Performing research on investigating influence of hydrogen on structure and properties of iron-based alloys using DFT methods (**WIEN2k**).

**Responsibilities:**

- Development of bash/Python utilities for monitoring WIEN2k calculations in PBS/Torque and SLURM queues and for processing data of calculations, developing mail notification system for the calculations.
- Installation of WIEN2k on a PC and on the university supercomputers (SKIF Ural, SKIF Aurora, Tornado).
- Teaching and consulting students how to use WIEN2k both on a PC and on supercomputers.

2007.06–2013.10 **QA Engineer, QA Team Leader**, *Applied Technologies Ltd., Chelyabinsk, Russia* (<https://www.appliedtech.ru/en/>), a partner of Rocket Software Inc., USA (<https://www.rocketsoftware.com>)



**Supervisor:** Valery Ermakov, CEO

**Software Projects:**

- Tivoli Enterprise Portal (TEP) (August 2011 – October 2013) — QA team leader. Responsibilities: software testing on Windows, Linux, Linux on z, z/OS operating systems, GUI testing automation using [IBM Rational Functional Tester](#), creation of TEP Automated Testing System (TATS) for automatic data verification using Python, REST API, JSON, XML, PHP, HTML, JavaScript, jQuery in six TEP-enabled products:
  - [Tivoli Advanced Reporting and Management for DFSMSHsm](#);
  - [Tivoli Advanced Audit for DFSMSHsm](#);
  - [Tivoli Advanced Catalog Management for z/OS](#);
  - [Tivoli Advanced Backup and Recovery for z/OS](#);
  - [Tivoli Advanced Allocation Management](#);
  - [Tivoli Automated Tape Allocation Manager for z/OS](#).
 Regular voice and video conversations with colleagues from the US.
- [Tivoli Storage Manager for z/OS Media](#) (February 2011 – August 2011) — QA engineer. Responsibilities: software testing on Windows, Linux on z, Solaris, AIX, z/OS operating systems with IBM DB2 databases, testing automation using IBM-developed tools, bash, batch.
- [IBM Tivoli Advanced Allocation Management](#) (January 2009 – January 2011) — QA engineer. Responsibilities: software testing on z/OS, testing automation using bash, JCL, REXX. Regular voice and video conversations with colleagues from the US.
- [Rocket Servergraph Data Protection Expert](#) (June 2007 – December 2008) — QA engineer. Responsibilities: software testing on Windows, Linux, Solaris, HP-UX, AIX operating systems with PostgreSQL databases, GUI testing automation using Autolt3, automation of data verification using Perl and SQL. Regular voice and video conversations with colleagues from the US.

2006–2007 **System administrator**, *Department of General and Theoretical Physics, South Ural State University, Chelyabinsk, Russia* (<https://www.susu.ru/en>)  
 Administer, secure and support Windows and Linux systems. Support HTTP server (Apache), proxy server (Squid). Help others to troubleshoot operating system, software, hardware and other issues. Also support the testing system for students.

## Education

2008.10–2012.09 **Ph.D. in Condensed Matter Physics (defended on September 19, 2012)**  
 *South Ural State University (National Research University), Chelyabinsk, Russia*  
*Thesis:* Study of impurities influence on the hydrogen dissolution energy in the bcc iron  
*Scientific adviser:* Prof. A.A. Mirzoev, Dr. of Sciences

2006.09–2008.06 **M.S. in Applied Mathematics and Physics (June 13, 2008)**

*South Ural State University (SUSU), Chelyabinsk, Russia*

*Thesis:* Computer simulation of influence of structural relaxation and impurities on dissolution energy of H in Fe

*Scientific adviser:* Prof. A.A. Mirzoev, Dr. of Sciences

*GPA:* 3.85 / 4.0

2002.09–2006.06 **B.S. in Applied Mathematics and Physics (June 20, 2006), *summa cum laude***

*South Ural State University (SUSU), Chelyabinsk, Russia*

*Thesis:* Binding energy of hydrogen in bcc iron lattice

*Scientific adviser:* Prof. A.A. Mirzoev, Dr. of Sciences

*GPA:* 3.69 / 4.0

## Honors, awards and certificates

- 2024.09 Spotlight Award in recognition of exceptional job performance (Brookhaven National Lab, Upton, NY)
- 2022.11 Engineering Award for distinguished contributions to Brookhaven National Laboratory's engineering objectives (a member of a team) — one of Brookhaven Lab's pinnacle awards (Brookhaven National Lab, Upton, NY). [news article](#)
- 2021.02 Certificate of Recognition for scientific contribution to the U.S. Department of Energy National Virtual Biotechnology Laboratory Project "Molecular Design and Analysis to Inform Therapeutics Related to COVID-19" (part of the DOE COVID-19 Pandemic Response in 2020)
- 2021.01 Certificate of Completion of the course "Scientific & Numeric Computing with Python" (James Powell, DUTC)
- 2021.01 Certificate of Completion of the course "Expert Level Python" (James Powell, DUTC)
- 2020 Spotlight Award in recognition of exceptional job performance (Brookhaven National Lab, Upton, NY)
- 2018 Spotlight Award in recognition of exceptional job performance (Brookhaven National Lab, Upton, NY)
- 2012 Certificate of summer school on computer simulations and massive calculations in modern physics with invited lectors from the US (Chelyabinsk, Russia)
- 2010 Certificate of summer school on computational material sciences (San-Sebastian, Spain)
- 2010 Scholarship of the University President (Chelyabinsk, Russia)
- 2009 Certificate of courses on high-performance computational systems (Nizhny Novgorod, Russia)
- 2009–2010 State contract (grant) on Federal Programme "Scientific and pedagogical staff for an innovative Russia" (Moscow, Russia)
- 2007 Diploma of the 13<sup>th</sup> All-Russian Physics Students Conference for the talk "Computer simulation of hydrogen atom in bcc iron" (Rostov-on-Don, Russia)
- 2006 Summa cum laude in Bachelor's studies (Chelyabinsk, Russia)
- 2006 Scholarship of the President of Russia

## Languages

English Full working proficiency, fluent  
Russian Native language

## Computer skills

**Data analysis, visualization** Python, NumPy, SciPy, Matplotlib, Bokeh, D3.js, Matlab/Octave, OriginPro, gnuplot, VESTA, XCrysDen, STM4, P4VASP, Molden, VMD, RasTop, etc.

**Programming** Python (including PyQT, NumPy, SciPy, Matplotlib, PIL), C++, Fortran, Matlab/Octave, JavaScript, jQuery, AngularJS, some React, WebGL, PHP, HTML5, bash, csh/tcsh, make, Perl, Autolt3, REXX, SQL, JCL

**Parallelization** MPI, OpenMP, Dask

**IDE/editors** VSCode, PyCharm, Eclipse, vim, some emacs

**Writing** L<sup>A</sup>T<sub>E</sub>X, BibT<sub>E</sub>X, TeXlipse/Eclipse, plasTeX, JabRef, MS Word

**OS** Linux (CentOS, SuSE, RedHat, Fedora, Ubuntu, Debian, Raspbian, etc.), Mac OS X, AIX, Solaris, HP-UX, Windows, DOS, z/OS

**Virtualization** Docker, Vagrant, VirtualBox, VMware

**VCS** GitHub, Git, SVN, CVS, Bazaar, Trac

**Atomistic simulation** VASP, WIEN2k, Quantum Espresso, CASTEP, SIESTA, CP2K, FHI-aims, QuantumWise ATK, Tinker, GULP, LAMMPS, DMACRYS, Phonopy, TB-LMTO-ASA

## References



Colleague

**Thomas Caswell**, Ph.D., Data Acquisition and Detectors Deputy Group Leader,  
DSSI, NSLS-II, Brookhaven National Laboratory, Upton, NY 11973  
[tcaswell@bnl.gov](mailto:tcaswell@bnl.gov) 📞 +1 (631) 344-3146



Colleague

**Stuart Campbell**, Ph.D., Chief Cyber Security Data Scientist,  
Information Technology Division, Brookhaven National Laboratory, Upton, NY 11973  
[scampbell@bnl.gov](mailto:scampbell@bnl.gov) 📞 +1 (631) 344-5578

*Additional references available upon request.*

## Publications

62. Z. Lentz, R. Tang-Kong, M. Rakitin, T. Morris, and S. Miskovich, "ML-assisted beamline optimization at LCLS," in *Proc. 20th Int. Conf. Accel. Large Exp. Phys. Control Syst. (ICALEPCS'25)*, ser. International Conference on Accelerator and Large Experimental Physics Control Systems, no. 20. JACoW Publishing, Geneva, Switzerland, Sep. 2025, paper FRAG001, pp. 1873–1877. <https://doi.org/10.18429/JACoW-ICALEPCS2025-FRAG001>
61. D. Gavrilov, T. Caswell, A. Sligar, and M. Rakitin, "Bluesky Queue Server for beamline control at NSLS-II," in *Proc. 20th Int. Conf. Accel. Large Exp. Phys. Control Syst. (ICALEPCS'25)*, ser. International Conference on Accelerator and Large Experimental Physics Control Systems, no. 20. JACoW

- Publishing, Geneva, Switzerland, Sep. 2025, paper THPD053, pp. 1706–1710. <https://doi.org/10.18429/JACoW-ICALEPCS2025-THPD053>
60. Y. Hidaka, D. Allan, and M. Rakitin, “A new Python middle layer framework: Particle Accelerator MIddle LAYer (PAMILA),” in *Proc. 6th North American Particle Accel. Conf. (NAPAC2025)*, ser. North American Particle Accelerator Conference, no. 2025. JACoW Publishing, Geneva, Switzerland, Aug. 2025, paper MOP005, pp. 48–51. <https://doi.org/10.18429/JACoW-NAPAC2025-MOP005>
  59. H. Wijesinghe, A. Barbour, L. Wiegart, E. Carlin, J. Einstein-Curtis, P. Moeller, R. Nagler, R. O’Rourke, N. Cook, and M. Rakitin, “Bluesky and Raydata: An Integrated Platform for Adaptive Experiment Orchestration,” in *SC24-W: Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis*, 2024, pp. 2162–2167. <https://doi.org/10.1109/SCW63240.2024.00271>
  58. A. Tayal, D. S. Coburn, D. Abel, M. Rakitin, O. Ivashkevych, J. Wlodek, D. Wierzbicki, W. Xu, E. Nazaretski, E. Stavitski, and D. Leshchev, “Five-analyzer Johann spectrometer for hard X-ray photon-in/photon-out spectroscopy at the Inner Shell Spectroscopy beamline at NSLS-II: design, alignment and data acquisition,” *Journal of Synchrotron Radiation*, vol. 31, no. 6, pp. 1609–1621, Nov. 2024. <https://doi.org/10.1107/S1600577524009342>
  57. T. W. Morris, M. Rakitin, Y. Du, M. Fedurin, A. C. Giles, D. Leshchev, W. H. Li, B. Romasky, E. Stavitski, A. L. Walter, P. Moeller, B. Nash, and A. Islegen-Wojdyla, “A general Bayesian algorithm for the autonomous alignment of beamlines,” *Journal of Synchrotron Radiation*, vol. 31, no. 6, pp. 1446–1456, Nov. 2024. <https://doi.org/10.1107/S1600577524008993>
  56. H. Goel, O. Chubar, R. Li, L. Wiegart, M. Rakitin, and A. Fluerasu, “Efficient end-to-end simulation of time-dependent coherent X-ray scattering experiments,” *Journal of Synchrotron Radiation*, vol. 31, no. 3, pp. 517–526, May 2024. <https://doi.org/10.1107/S1600577524001267>
  55. N. M. Cook, A. M. Barbour, E. G. Carlin, J. A. Einstein-Curtis, R. Nagler, R. O’Rourke, M. Rakitin, L. Wiegart, and H. Wijesinghe, “Integrating Online Analysis with Experiments to Improve X-Ray Light Source Operations,” in *Proc. 19th Int. Conf. Accel. Large Exp. Phys. Control Syst. (ICALEPCS’23)*, ser. International Conference on Accelerator and Large Experimental Physics Control Systems, no. 19. JACoW Publishing, Geneva, Switzerland, Feb. 2024, paper TUSDSC02, pp. 921–924. <https://doi.org/10.18429/JACoW-ICALEPCS2023-TUSDSC02>
  54. J. A. Einstein-Curtis, D. T. Abell, Y. Du, A. Giles, M. V. Keilman, J. Lynch, P. Moeller, T. Morris, B. Nash, I. V. Pogorelov, M. Rakitin, and A. L. Walter, “Online Models for X-ray Beamlines Using Sirepo-Bluesky,” in *Proc. 19th Int. Conf. Accel. Large Exp. Phys. Control Syst. (ICALEPCS’23)*, ser. International Conference on Accelerator and Large Experimental Physics Control Systems, no. 19. JACoW Publishing, Geneva, Switzerland, Feb. 2024, paper MO3BCO05, pp. 165–170. <https://doi.org/10.18429/JACoW-ICALEPCS2023-MO3BCO05>
  53. P. M. Maffettone, D. B. Allan, A. Barbour, T. A. Caswell, D. Gavrilov, M. D. Handwell, T. Morris, D. Olds, M. Rakitin, S. I. Campbell, and B. Ravel, *Book “Methods and*

*Applications of Autonomous Experimentation". Chapter 8: "Artificial Intelligence Driven Experiments at User Facilities", 1st ed. Chapman & Hall/CRC Computational Science, 2023, ch. Chapter 8. <https://doi.org/10.1201/9781003359593>*

52. M. Rakitin, R. Bode, T. W. Morris, A. C. Giles, A. L. Walter, J. K. Lynch, J. Maldonado, Y. Du, B. Romasky, M. Fedurin, P. Moeller, and B. Nash, "Recent updates of the Sirepo-Bluesky library for virtual beamline representation," in *Advances in Computational Methods for X-Ray Optics VI*, O. Chubar and T. Tanaka, Eds., vol. 12697, International Society for Optics and Photonics. SPIE, 2023, p. 126970D. <https://doi.org/10.1117/12.2678030>
51. T. W. Morris, Y. Du, M. Fedurin, A. C. Giles, P. Moeller, B. Nash, M. Rakitin, B. Romasky, A. L. Walter, N. Wilson, and A. Wojdyla, "Latent Bayesian optimization for the autonomous alignment of synchrotron beamlines," in *Advances in Computational Methods for X-Ray Optics VI*, O. Chubar and T. Tanaka, Eds., vol. 12697, International Society for Optics and Photonics. SPIE, 2023, p. 126970B. <https://doi.org/10.1117/12.2677895>
50. B. Nash, M. S. Rakitin, D. T. Abell, M. Keilman, P. Moeller, I. Pogorelov, Y. Du, A. Giles, J. Lynch, T. W. Morris, A. L. Walter, and N. Goldring, "Reduced model representations of synchrotron radiation and a software framework for beamline control," in *Advances in Computational Methods for X-Ray Optics VI*, O. Chubar and T. Tanaka, Eds., vol. PC12697, International Society for Optics and Photonics. SPIE, 2023, p. PC1269703. <https://doi.org/10.1117/12.2676921>
49. L. Huang, T. Wang, O. Chubar, G. Dovillaire, A. He, M. Rakitin, Y. Yang, A. M. Kiss, and M. Idir, "Investigation of x-ray Hartmann wavefront sensing: from simulation to the initial experiment test," in *Advances in Computational Methods for X-Ray Optics VI*, O. Chubar and T. Tanaka, Eds., vol. PC12697, International Society for Optics and Photonics. SPIE, 2023, p. PC1269705. <https://doi.org/10.1117/12.2675754>
48. H. Goel, O. Chubar, L. Wiegart, A. Fluerasu, R. Li, A. He, M. Rakitin, M. Lin, P. Moeller, and R. Nagler, "GPU accelerated simulations of time-dependent coherent x-ray scattering experiments," in *Advances in Computational Methods for X-Ray Optics VI*, O. Chubar and T. Tanaka, Eds., vol. 12697, International Society for Optics and Photonics. SPIE, 2023, p. 1269709. <https://doi.org/10.1117/12.2677888>
47. N. Cook, E. Carlin, R. Nagler, R. O'Rourke, A. Barbour, M. Rakitin, and L. Wiegart, "An online analysis platform for improving x-ray light source operations," in *Proc. IPAC'23*, ser. IPAC'23 - 14th International Particle Accelerator Conference, no. 14. JACoW Publishing, Geneva, Switzerland, May 2023, paper THPA128, pp. 4266–4268. <https://doi.org/10.18429/JACoW-IPAC2023-THPA128>
46. M. Rakitin, S. Campbell, D. Allan, T. Caswell, D. Gavrilov, M. Hanwell, and S. Wilkins, "Next generation experimental data access at NSLS-II," *Journal of Physics: Conference Series*, vol. 2380, no. 1, p. 012100, Dec. 2022. <https://doi.org/10.1088/1742-6596/2380/1/012100>
45. B. Nash, D. Abell, R. Nagler, P. Moeller, M. Keilman, I. Pogorelov, N. Goldring, M. Rakitin, J. Lynch, A. Giles, A. Walter, J. Maldonado, T. Morris, S. Bak, and Y. Du, "Combining diagnostics, modeling, and control systems for automated alignment of the

- TES beamline,” *Journal of Physics: Conference Series*, vol. 2380, no. 1, p. 012103, Dec. 2022. <https://doi.org/10.1088/1742-6596/2380/1/012103>
44. H. Goel, O. Chubar, L. Wiegart, A. Fluerasu, R. Li, A. He, M. Rakitin, P. Moeller, and R. Nagler, “Developments in SRW Code and Sirepo Framework Supporting Simulation of Time-Dependent Coherent X-ray Scattering Experiments,” *Journal of Physics: Conference Series*, vol. 2380, no. 1, p. 012126, Dec. 2022. <https://doi.org/10.1088/1742-6596/2380/1/012126>
  43. T. W. Morris, M. Rakitin, A. Giles, J. Lynch, A. L. Walter, B. Nash, D. Abell, P. Moeller, I. Pogorelov, and N. Goldring, “On-the-fly optimization of synchrotron beamlines using machine learning,” in *Optical System Alignment, Tolerancing, and Verification XIV*, J. Sasián and R. N. Youngworth, Eds., vol. 12222, International Society for Optics and Photonics. SPIE, 2022, p. 122220M. <https://doi.org/10.1117/12.2644996>
  42. L. Huang, T. Wang, O. Chubar, G. Dovillaire, A. He, M. Rakitin, and M. Idir, “Simulation of X-ray Hartmann wavefront sensing with the Synchrotron Radiation Workshop,” *Opt. Express*, Oct. 2022. <https://doi.org/10.1364/oe.470197>
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## Conferences, talks, workshops and schools

- 2024.09 **Invited talk** — *Automated Beamline Alignment and Experiment Steering* , SSRL/LCLS Users' Meeting, "MFX-HE: Opportunities for High-Throughput Multimodal Structural Studies and Their Computational Challenges" Workshop, Menlo Park, CA
- 2024.09 **Workshop organizer** — *Bluesky and Bluesky-Queueserver tutorials* , EPICS Meeting, Oak Ridge National Laboratory, Oak Ridge, TN
- 2023.11 **Invited talk** — *Bluesky for Multimodal Studies at NSLS-II* , "Integrating acquisition and AI in tomography" Workshop, Leiden, The Netherlands
- 2023.08 *Recent updates of the Sirepo-Bluesky library for virtual beamline representation*, SPIE Optics + Photonics, Advances in Computational Methods for X-Ray Optics VI , San Diego, CA

- 2023.05 **Invited teacher** at *Beamline and Optics Modeling School* , Advanced Light Source, Berkeley, CA
- 2023.04 *Ophyd beyond EPICS* , NSLS-II & CFN Users' Meeting, Workshop 4 — “Scientific Python for Data Acquisition, Management, and Analysis”, Brookhaven National Laboratory, Upton, NY
- 2023.03 *AE124: Simulation-aided Instrument Optimization using Artificial Intelligence and Machine Learning Methods* , 25<sup>th</sup> Annual Accelerator Test Facility (ATF) Users' Meeting, Brookhaven National Laboratory, Upton, NY
- 2022.08 **Invited talk** — *Automated beamline alignment at NSLS-II* , ALS User Meeting, “Instrumentation and Computation for the Upgraded ALS” Workshop, Virtual Meeting
- 2022.05 **Invited talk** — *24/7 access to your virtual beamline with Sirepo* , NSLS-II & CFN Users' Meeting, Workshop 6 — “Data Access and Machine Learning at NSLS-II”, Brookhaven National Laboratory, Upton, NY
- 2022.03 **Invited talk** — *Next generation experimental data access at NSLS-II* , SRI2021, Virtual Conference
- 2020.02 **Invited talk** — *Bluesky Data Collection Framework*, Canadian Light Source, Saskatoon, Canada
- 2019.10 *Overview of Bluesky*, Imaging Workshop, Oak Ridge National Laboratory, Oak Ridge, TN
- 2019.07 *Kitware training on vtk.js, girder, tomviz, vtk, paraview, and cmake*, Clifton Park, NY
- 2019.06 *Jupyter Community Workshop* , LBNL, Berkeley, CA
- 2019.06 *Ptycho Developer Workshop* , LBNL, Berkeley, CA
- 2019.01 **Invited talk** — *Bluesky, ophyd, pseudo motors, detectors* , Automation in Beamline Control and Data Acquisition workshop, HZB, BESSY-II, Berlin, Germany
- 2019.01 **Invited tutorials on Sirepo and Bluesky** , Automation in Beamline Control and Data Acquisition workshop, HZB, BESSY-II, Berlin, Germany
- 2018.10 *NOBUGS 2018*  , Brookhaven National Laboratory, Upton, NY
- 2018.07 *SciPy 2018* , Austin, TX
- 2018.06 *EPICS Collaboration meeting* , APS, Argonne National Laboratory, Lemont, IL
- 2018.05 *2018 NSLS-II & CFN Users' Meeting: assisting with “Hands-On Data Acquisition and Analysis Tutorial”* , Brookhaven National Laboratory, Upton, NY
- 2017.11 *Sirepo — an open-source browser interface for X-ray source and optics simulations*, ORNL Visualization Hackathon, Oak Ridge National Laboratory, Oak Ridge, TN
- 2017.10 *Sirepo — an open-source cloud-based software interface for X-ray source and optics simulations*, NSLS-II Lunchtime seminar, Brookhaven National Laboratory, Upton, NY
- 2017.08 **Invited talk** — *Sirepo: a web-based interface for physical optics simulations — its deployment and use at NSLS-II*, SPIE Optical Engineering + Applications , San Diego, CA
- 2017.05 *2017 NSLS-II & CFN Users' Meeting* , Brookhaven National Laboratory, Upton, NY
- 2016.12 *Collaboration meeting with RadiaSoft LLC*, Boulder, CO
- 2016.12 *Early Career Researcher Symposium 2016*, Brookhaven National Laboratory, Upton, NY

- 2016.10 *Software for Optical Simulations (SOS) Workshop*  , ICTP, Trieste, Italy
- 2016.05 *2016 NSLS-II & CFN Joint Users' Meeting* , Brookhaven National Laboratory, Upton, NY
- 2015.11 *Sensitivity, Error and Uncertainty Quantification for Atomic, Plasma, and Material Data* , IACS, Stony Brook University, Stony Brook, NY
- 2015.10 *Collaboration Meeting on "Simulation and Modeling for SR Sources and X-Ray Optics"*, NSLS-II, Brookhaven National Laboratory, Upton, NY
- 2015.07 **Invited talk** — *Crystal and protein structure modeling, software development and applications*, Brookhaven National Laboratory, Upton, NY
- 2015.06 *Advances in Functional Materials — Conference 2015*, Stony Brook University, Stony Brook, NY
- 2015.06 *Recent progress in USPEX development*, Group seminar, Stony Brook University, Stony Brook, NY
- 2015.04 **Invited talk** — *Crystal structure prediction from first principles* , Humboldt-Universität zu Berlin, Institut für Physik, Berlin, Germany
- 2015.04 *Oracle R, Advanced & predictive Analytics Workshop*, Stony Brook University, Stony Brook, NY
- 2015.02 *MATLAB & Simulink for Project-Based Learning using LEGO MINDSTORMS EV3*, Stony Brook University, Stony Brook, NY
- 2015.01 *IACS workshop "Intro to Python"* , Stony Brook University, Stony Brook, NY
- 2014.11 *Novel phase of beryllium fluoride at high pressure*, Group seminar, Stony Brook University, Stony Brook, NY
- 2014.11 **Invited tutor** — *Theory and Computation for Interface Science and Catalysis: Fundamentals, Research and Hands on Engagement using VASP*  , Brookhaven National Laboratory, Upton, NY
- 2014.10 *MATLAB and Simulink Complimentary Technical Sessions at Stony Brook University*, Stony Brook, NY
- 2014.09 *Proteins structure prediction using USPEX*, Group seminar, Stony Brook University, Stony Brook, NY
- 2014.09 *USPEX tests for Tinker: Different amino-acids XYZ-20, ALA-40, CASP10, REMD*, Group seminar, Stony Brook University, Stony Brook, NY
- 2014.07 *Introduction to Python*, Group seminar, Stony Brook University, Stony Brook, NY
- 2014.04 *SiO<sub>2</sub> and BeF<sub>2</sub> phase transformation under pressure. Proteins simulation with Tinker interface for USPEX*, Group seminar, Stony Brook University, Stony Brook, NY
- 2014.03 *CECAM workshop "Simulation of biomolecular interactions with inorganic and organic surfaces as a challenge for future nanotechnologies"* , Toulouse, France
- 2014.03 *USPEX mini-workshop*, Group seminar, Stony Brook University, Stony Brook, NY
- 2014.02 *SiO<sub>2</sub> and BeF<sub>2</sub> phase transformation under pressure*, Group seminar, Stony Brook University, Stony Brook, NY
- 2013.11 *Study of impurities influence on the hydrogen dissolution energy in the bcc iron*, Group seminar, Stony Brook University, Stony Brook, NY

- 2013.08 *2<sup>nd</sup> summer school on computer simulations in modern physics* , Chelyabinsk, Russia
- 2012.08 *Summer school on computer simulations and massive calculations in modern physics* , Chelyabinsk, Russia
- 2011.10 *6<sup>th</sup> All-Russian scientific-technical conference "Physical properties of metals and alloys"* , Yekaterinburg, Russia
- 2011.09 *International conference "Thermodynamics 2011"*, Athens, Greece
- 2011.05 *2<sup>nd</sup> All-Russian youth school-conference "Modern problems of metal science"* , Pitsunda, Abkhazia
- 2010.07 *International symposium "Metal-hydrogen systems. Fundamentals and applications"*  , Moscow, Russia
- 2010.06 *International summer school "Computational Materials Science"* , San Sebastian, Spain
- 2010.03 *All-Russian conference "Parallel computing technologies 2010"* , Ufa, Russia
- 2010.02 *33<sup>rd</sup> International conference on theoretical physics "Kourovka-2010"*  , Novouralsk, Russia
- 2009.11 *9<sup>th</sup> International conference "High-performance parallel computing on cluster systems"* , Vladimir, Russia
- 2009.10 *12<sup>th</sup> V.A. Fock All-Russian conference on quantum and computational chemistry* , Kazan, Russia
- 2007.04 *13<sup>th</sup> All-Russian students conference in physics*  , Rostov-on-Don, Russia